Status	Finished
Started	Tuesday, 13 October 2026, 3:50 PM
Completed	Tuesday, 13 October 2026, 3:55 PM
Duration	4 mins 31 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Incorrect

Mark 0.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 99

b. 89 and 94

c. 94 and 99 ✗

d. 90 and 100

Question **2**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It can be applied only if the list is sorted ✓

b. It works on unsorted lists

c. It always starts from the beginning of the list

d. It compares elements sequentially

Question **3**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By using a hash function

b. By comparing each element sequentially

c. By dividing the list length by two ✓

d. By starting from the first element

Question **4**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

- a. O(n log n)
- b. O(log n)
- c. O🥎
- d. O(1) ✓

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Linear search ✓
- b. Hashing
- c. Dynamic search
- d. Binary search

Question **6**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

- a. When the item is the last element in the array
- b. When the item is not the array at all
- c. Item is the last element in the array or item is not there at all
- d. When the item is somewhere in the middle of the array ✓

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- a. Linear search can be applied to both sorted and unsorted lists. ✓
- b. Linear search requires the list to be sorted.
- c. Linear search divides the list into halves.
- d. Linear search is more efficient than binary search on large lists.

Question **8**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Hash search
b. Binary search
c. Linear search ✓
d. Interpolation search

Question **9**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Sorting elements in a list
b. Deleting elements from a list
c. Determining whether an element is present in a list ✓
d. Inserting elements into a list

Question **10**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and unsorted
b. When the list is small and sorted
c. When the list is small and unsorted ✓
d. When the list is large and sorted

Question **11**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search does not require dividing the list
b. Binary search can find multiple instances of the target element
c. Binary search has a lower time complexity on large, sorted lists ✓
d. Binary search works on unsorted lists

Question **12**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n-1
- b. log n
- c. n ✓
- d. n/2

Question **13**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. n/2
- b. n ✓
- c. 1
- d. log n

Question **14**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

- a. 90 and 94
- b. 89 and 94
- c. 89 and 99
- d. 90 and 99 ✓

Question **15**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. At the beginning of the list
- b. In the middle of the list
- c. In the left sublist
- d. In the right sublist ✓